Analysis Report Template

Student Full Name		Student ID No.	

Case Study Title
Case Study (A): Predicting Clients Loan Status. **[Total 43 Marks]**
Research Question: Does machine learning have the potential to assist banker and finance analysts in predicting which client can be approved a loan?

Task (1) – Domain Understanding: Classification

The financial risk analysts decided that classification modelling is required. Indicate in the table below for each of the listed variables in your data which ones you should RETAIN and can be included in the classification modelling of loan approval status, should DROP (REMOVE). Justify your decision logically and/or by research (for any research, include in-text citation using the Harvard style) **[3 Marks]**

Variable Name	RETAIN or DROP	Brief justification for retention or dropping with in-text citation.
ID		
Sex		
Age		
Education Qualifications		
Income		
Home Ownership		
Employment Length		
Loan Intent		
Loan Amount		
Loan Interest Rate		
Loan-to-Income Ratio (LTI)		
Payment Default on File		
Credit History Length		
Loan Approval Status		
Maximum Loan Amount		

Credit Application Acceptance		
References		

Task (2) – Exploring and Understanding Your Dataset

With the aid of your Final Python Notebook 1, for your RETAINED input variables and your class "Target" variable, produce 1)**basic descriptive stat**s and 2)**variable scale type, then** 3)p**lot the distribution of your target variable**. (Paste the three screenshots of code OUTPUTS ONLY for evidence of these elements).

[2 Marks]

Task (3) – Data Preparation: Cleaning and Transforming your data

a) With the aid of your Final Python Notebook 1, when you first explored your retained variables in the loan dataset, you may have found some issues. **1) Report any issues you found in your retained dataset variables**. Based on the issues you found in your data, **2)suggest a suitable possible method to fix each of these issues** and **3)provide your justification for using your suggested fix method**. Use the table below to organise your findings and analysis, and add more rows if needed

[4 Marks]

	Variable Name	Issue found	Proposed fix	Justification for used fix method
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
⋮	⋮	⋮	⋮	⋮

Task (3) – Data Preparation: Cleaning and Transforming your data

b) With the aid of Python packages and your Final Python Notebook 1, implement your suggested fixes of issues in the previous Task 3-a in your final Python Notebook1.

1) Show evidence (before and after) of implementing your suggested fix to the problems you identified for your dataset in Task 3-a.

To show your evidence, paste screenshots of your relevant code OUTPUTS ONLY (Do not paste the code).

2) Indicate and annotate the issue and the fix in each of your provided evidence screenshots.

[4 Marks]

Output screenshot of the issue before the fix	Output screenshot after fixing the issue

Output screenshot of the issue before the fix	Output screenshot after fixing the issue

Task (4) – Classification Modelling of Clients Loan Approval Status

a) In your Final Python Notebook 2, you built THREE different models to predict loan approval status: Logistic Regression (LR), K Nearest Neighbour (KNN) and Naïve Bayes (NB). These algorithms are a mix of parametric and non-parametric algorithms.
1) Note down the type of each algorithm (parametric vs non-parametric),
2) name any learnable parameters, and
3) list any strategic hyperparameters for each algorithm which you want to consider tuning. Organise your answer in the table below:

[3 Marks]

Algorithm Name	Algorithm Type	Learnable Parameters	Some Strategic Hyperparameters
Naïve Bayes (NB)			
Logistic Regression (LR)			
K-Nearest Neighbour (KNN)			

Task (4) – Classification Modelling of Clients Loan Approval Status

b) With the aid of your Final Python Notebook 2**,** use the training–test split approach with your retained applicable input features only and the target output feature to build your predictive classification models.

[3 Marks]

i. Screenshot
1) the list of all feature names used for building your classification models and the corresponding
2) data shape function output. (Paste screenshots of the relevant code output only; do not paste the Python code).

ii. In less than 150 words, **research and justify (defend) your choice of the training-test split ratio** and provide an in-text citation.

References

iii. Provide as evidence **the code block line** and **code output from** your Final Python Notebook 2 that ensures two conditions:
1) all your models were tested on the same test instances (clients) in your dataset;
2) the labels ratio of approval Status "Accept" to "Reject" is the same in the training and test subsets.
3) State the training-test split function parameters in your code line that are responsible for meeting both conditions.
4) In less than 150 words**, research and justify (defend) your decision to implement both conditions** in your Python Notebook 2 notebook with in-text citations where possible.

References with in-text citation

Task (5) – Evaluating your Loan Approval Status Classification Models

Your finance team provided the following success criteria to guide you when evaluating and selecting your best model: ***"When evaluating your model's performance, which addresses your first research question (a). The model is expected to misclassify clients' applications. Thus, the model should aim to predict as many "Reject" status for clients as many as possible to decrease the risk of future defaulted loan payments. However, the model should demonstrate that its high "Reject" prediction rate is mainly due to a larger number of correctly detected (predicted) rejected loan applications."***
a) With the aid of Final Python Notebook 2, for each of your models (Logistic Regression LR, Naive Bayes NB and K-Nearest Neighbours KNN) **1) paste the test confusion matrix**, **2) the classification report and** **3) the AUC-ROC curve graphs** as screenshots from the output of your Python code. **[3 Marks]**

1) Screenshot of the test confusion matrix for (NB, LR, and KNN); make sure you title each matrix with its algorithm name

2) Screenshot of the classification report for (NB, LR, and KNN); make sure you title each classification report with its algorithm name

3) Screenshot of the AUC-ROC Curve for (NB, LR, and KNN); make sure you title each graph with its algorithm name

Task (5) – Evaluating your Loan Approval Status Classification Models

b) Five different classification evaluation metrics are calculated in your Final Python Notebook 2.
1) State which evaluation metric/metrics to "USE or "DO NOT USE" to closely interpret the above success criteria.
2) For justification, explain how closely your choice of "USE" or "DO NOT USE" for a metric interprets the given success criteria.
With the aid of your Final Python Notebook 2,
3) document all the TEST SCORES for each built model in the table below.

[7 Marks]

Metric	1) USE or DO NOT USE	2) Justification for choosing "USING" or "NOT USING" this metric in relation to the success criteria	Model	3) Metric Test Score
Accuracy			NB	
			LR	
			KNN (k=?)	
Recall			NB	
			LR	
			KNN (k=?)	
Precision			NB	
			LR	
			KNN (k=?)	
F1-score			NB	
			LR	
			KNN (k=?)	
AUC-Roc			NB	
			LR	
			KNN (k=?)	

Task (5) – Evaluating your Loan Approval Status Classification Models

c) **Suggest a single best approval status classification model** based on the 'USED' performance metrics scores you identified in Task (5-b). In less than 100 words, briefly **describe how well your best model satisfies the needs of your finance team.**

[2 Marks]

Task (5) – Evaluating your Loan Approval Status Classification Models

d) To enhance your selected best model/s performance (from Task 5-c), tune some of its possible hyperparameters, which you indicated in Task (4-a) for that specific algorithm. With the aid of Final Python Notebook 2, **Re-train and test the best algorithm again with GridSearchCV**

[5 Marks]

i. With the aid of your Final Python Notebook 2,

1) Paste into this report the line of code which shows evidence of specifying a parameters grid and applying the GridSearchCV function to rebuild your selected best model.

2) Then, document the estimated best hyperparameters for the optimised model.

ii. With the aid of your Final Python Notebook 2,

1) paste into this report the test confusion matrix for your best model before and after hyperparameter tuning.

2) Also, document the new score/s of the "USED" performance metric/s of your choice to interpret the success criteria indicated in Task (5.b) before and after tuning.

3) Comment on whether the tuning of hyperparameters of your best model improved its positive predictive ability in line with the success criteria.

Task (5) – Evaluating your Loan Approval Status Classification Models

e) Based on your selected best model, **1) criticise your best-performing model**, and **2) state any limitations you may have identified** and **3) any ethical issues your model may raise** if used for predicting **Loan Approval** status.

[2 Marks]

Task (5) – Evaluating your Loan Approval Status Classification Models

f) With the aid of your Final Python Notebooks 3**,** combine only TWO out of the THREE base learners (NB, LR, KNN) that you already built into a probability-based voting ensemble classifier.

[5 Marks]

i. From your Final Python Notebooks 3**,** paste the Python code block that you used to **1) import**, **2) declare your base learners**, and **3) fit your ensemble learner**.

ii. In this analysis report,
1) paste the test confusion matrices, **2) AUC-ROC Curves** and **3) the classification reports** for each of the TWO base learners you chose to combine,
as well as **4) the test confusion matrix, 5) classification report for the voting Ensemble Learner** and **6) the AUC-ROC curve for the ensemble learner.**
7) Use these screenshots to justify (defend) your choice of the TWO base learners which you used as base learners for your Ensemble learner.

1) Paste the test confusion matrices for each base learner (2 x matrices)

2) Paste the AUC-ROC for each Base learner (2 x AUC-ROC graphs)

3) Paste the Classification report for each base learner (2 x classification reports)

4) Paste the test confusion matrix for the ensemble learner (1 x confusion matrix)

5) Paste the AUC-ROC for the ensemble learner (optional) (1 x AUC-ROC graph)

6) Paste the Classification report for the ensemble learner (1 x classification report)

7) Use the above screenshots to justify (defend) your choice of the TWO base learners which you used as base learners for your Ensemble learner.

iii. Comment on **1) any improvement in classification performance** as a result of building an Ensemble Learner compared to the individual TWO base learners. **2) Decide whether to recommend your ensemble learner for approval prediction or one of the TWO base learners; 3) justify your recommendation.**

Case Study Title

Case Study (B): Predicting Maximum Loan Amount.
Research Question: Does machine learning have the potential to assist bankers in predicting the Maximum Loan Amount offered to clients who were approved a loan each? [Total 36 Marks]

Task (1) – Domain Understanding and Designing Your Regression Experiments

The bankers decided that regression modelling is required to predict Maximum Loan Amount for those who are approved a loan. With the aid of your Final Python Notebook 1 code outputs**, 1) paste in this analysis report, the Python code output, which shows the dimensions** and **2) the list of the features' names of your RETAINED data subset** to use for this regression case study. **[2 Marks]**

Task (2) – Modelling: Build Predictive Regression Models

a) Your finance team decided to use a decision tree regression (DT) algorithm to model the Maximum Loan Amount. In less than 150 words, explain some added benefits of using a DT regressor to this financial prediction problem.

[2 Marks]

References with in-text citation

Task (2) – Modelling: Build Predictive Regression Models

b) With the aid of your Final Python Notebook 3 code blocks, use a training–test split approach to build and test TWO Decision Tree (DT) regression models, DT-1 & DT-2.

[6 Marks]

i. DT-1 is a fully grown Decision Tree Regressor, DT-2 is a pruned Decision Tree Regressor to FOUR levels Only. **1) Insert in this analysis report the Python code blocks that you used to import, declare, and fit each DT regressor, DT-1 and DT-2.**

ii. Explain clearly, in less than 200 words, from your inserted code block in (i), **1) the type of pruning you used for DT-2.**
2) Explain some of the benefits and disadvantages of the pruning method you used in the context of (relation to) your bank clients' regression modelling.

Task (2) – Modelling: Build Predictive Regression Models

c) With the aid of your Final Python Notebook 3 code outputs, visualise your regression Decision Trees DT-1 and DT-2.
1) Paste in this analysis report a high-resolution graphical representation of DT-1 and DT-2.

[4 Marks]

Task (3) – Evaluating your Maximum Loan Amount DT Regression Models

Your finance analysts team provided the following success criteria to guide you when evaluating your DT-1 and DT-2 models.

"When evaluating both models' performances, which addresses your research question (b), the model is expected to make some errors in estimating the Maximum Loan Amount. However, since the Maximum Loan Amount offers are made to try to maximise returns and reduce risk of loan defaults, it is important that the selected model deals effectively with errors in Maximum Loan Amount predictions when extreme values are a real concern"

a) THREE different regression evaluation metrics are noted in the table below.
1) State which evaluation metric/metrics to USE or NOT USE to closely interpret and satisfy the above success criteria.
2) Justify (Defend) your choice of USE or DO NOT USE for each metric.
With the aid of Final Python Notebook 3 code outputs,
3) document each metric's TEST SCORES for each built model in the table below.

[8 Marks]

Metric	1) USE or DO NOT USE	2) Justification for choosing "USING" or "NOT USING" this metric in relation to the success criteria	Model	3) Metric Test Score
MSE			DT-1 (Fully Drown)	
			DT-2 (Pruned)	
MAE			DT-1 (Fully Drown)	
			DT-2 (Pruned)	
R-Square			DT-1 (Fully Drown)	
			DT-2 (Pruned)	

Task (3) – Evaluating your Maximum Loan Amount DT Regression Models

b) **1) Suggest a single best regression model (DT-1 or DT-2)** based on your 'USED' performance metric/s scores, which you defended in Task (3a). **2) Explain how your suggested model fulfils the success criteria.**

[4 Marks]

Task (3) – Evaluating your Maximum Loan Amount_DT Regression Models

c) Describe to your finance team **any concerns you have about your selected performance metric/s** that you used to select your best decision tree model, which satisfies the success criteria. [200 words maximum] with in-text citation.

[4 Marks]

References with in-text citation

Task (3) – Evaluating your Maximum Loan Amount_DT Regression Models

d) Client 60256 was Approved a Loan. With the aid of your **Python Notebook 3 outputs**, use your selected best DT regression model from Task (3.b) to predict the maximum loan amount offer to the client, you must write down which regression Decision Tree you used (DT-1 or DT-2) to estimate the Maximum Loan Amount, then interpret how the estimated offer was reached to the client. Client 60256 attributes' values are in the following table: **[6 Marks]**

Variable Name	Value
ID	**60256**
Sex	**Female**
Age	**56 Years old**
Education Qualifications	**Unknown**
Income	**57,000**
Home Ownership	**Rent**
Employment Length	**15**
Loan Intent	**Medical**
Loan Amount	**25700**
Loan Interest Rate	**23%**
Loan-to-Income Ratio (LTI)	**10%**
Payment Default on File	**No**
Credit History Length	**35**
Loan Approval Status	**Approved**
Maximum Loan Amount	**?**
Credit Application Acceptance	**0**